

Лабораторная работа №7

Адресация IPv4 и IPv6. Настройка DHCP

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Информация

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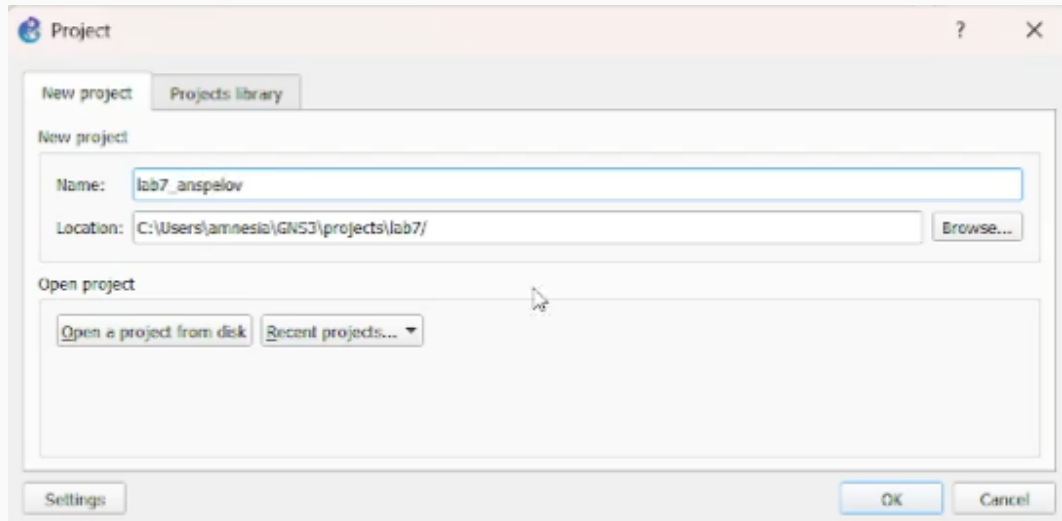
Вводная часть

- Целью данной работы является получение навыков настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.

Основная часть

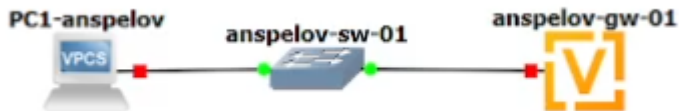
Настройка DHCP в случае IPv4

- Создание нового проекта в GNS3.



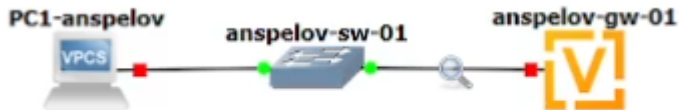
Настройка DHCP в случае IPv4

- Размещение устройств в соответствии с топологией в лабораторной работе. Изменение названий устройств.



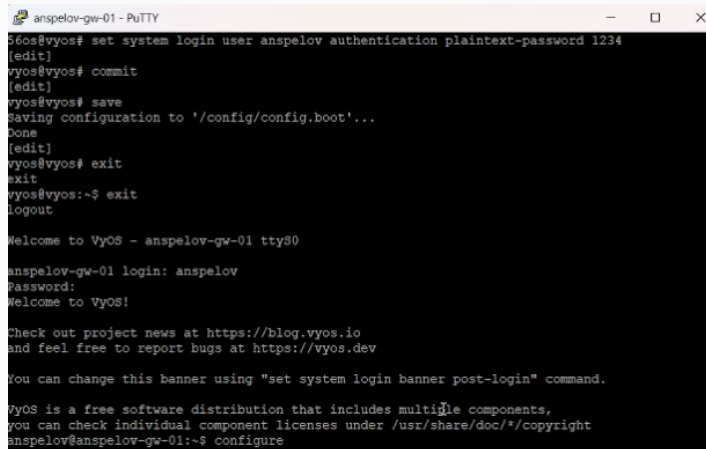
Настройка DHCP в случае IPv4

- Включение захвата трафика.



Настройка DHCP в случае IPv4

- Настройка образа VyOS и переход на маршрутизаторе в режим конфигурирования, изменение имени устройства и доменного имени, замена системного пользователя.



```
anspelov-gw-01 - PuTTY
56os@vyos# set system login user anspelov authentication plaintext-password 1234
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ exit
logout

Welcome to VyOS - anspelov-gw-01 ttyS0

anspelov-gw-01 login: anspelov
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
anspelov@anspelov-gw-01:~$ configure
```

Настройка DHCP в случае IPv4

- Настройка адресации IPv4, добавление конфигурации DHCP-сервера на маршрутизаторе.

```
an-name ansplov.net1# set service dhcp-server shared-network-name ansplov do
edit]
e-server 10.0.0.1gw-01# set service dhcp-server shared-network-name ansplov na
edit]
net 10.0.0.0/24 default-router 10.0.0.1p-server shared-network-name ansplov su
edit]
server 10.0.0.1gw-01# set service dhcp-server shared-network-name ansplov name

Configuration path: service dhcp-server shared-network-name ansplov [me-server] is not valid
Set failed

edit]
edit]
ansplov@ansplov-gw-01#
edit]
net 10.0.0.0/24 range hosts start 10.0.0.2server shared-network-name ansplov su
edit]
ansplov@ansplov-gw-01# set service dhcp-server

Configuration path: [service dhcp-server] already exists

edit]
ansplov@ansplov-gw-01# set service dhcp-server

Configuration path: [service dhcp-server] already exists

edit]
ansplov@ansplov-gw-01# set service dhcp-server

Configuration path: [service dhcp-server] already exists

edit]
ansplov@ansplov-gw-01# set service dhcp-server

Configuration path: [service dhcp-server] already exists

edit]
net 10.0.0.0/24 range hosts stop 10.0.0.253rver shared-network-name ansplov su
edit]
```

Настройка DHCP в случае IPv4

- Просмотр статистики DHCP-сервера и выданных адресов.

```
delov@anspelov-gw-01:~$ show dhcp server statistics
.      Size    Leases    Available    Usage
-----
delov   252      0          252    0%
delov@anspelov-gw-01:~$ show dhcp server leases
address    Hardware address    State    Lease start    Lease expiration    Remaining    Pool    Ho
me
-----
```

Настройка DHCP в случае IPv4

- Настройка оконечного устройства PC1-anspelov.

```
PC1-anspelov - PuTTY
Opcode: 2 (REPLY)
Client IP Address: 0.0.0.0
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Offer
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = anspelov.net

Opcode: 1 (REQUEST)
Client IP Address: 10.0.0.2
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Request
Option 54: DHCP Server = 10.0.0.1
Option 50: Requested IP Address = 10.0.0.2
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00
Option 12: Host Name = PC1-anspelov1

Opcode: 2 (REPLY)
Client IP Address: 10.0.0.2
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Ack
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = anspelov.net

ip 10.0.0.2/24 GW 10.0.0.1
```

Настройка DHCP в случае IPv4

- Проверка конфигурации IPv4 на узле и пинг маршрутизатора.

```
PC1-anspelov> show ip
NAME           : PC1-anspelov[1]
IP/MASK        : 10.0.0.2/24
GATEWAY        : 10.0.0.1
DNS            : 10.0.0.1
DHCP SERVER    : 10.0.0.1
DHCP LEASE     : 86362, 86400/43200/75600
DOMAIN NAME    : anspelov.net
MAC            : 00:50:79:66:68:00
LPORT         : 10003
RHOST:PORT     : 127.0.0.1:10004
MTU:           : 1500

PC1-anspelov> ping 10.0.0.1 -c 2
84 bytes from 10.0.0.1 icmp_seq=1 ttl=64 time=7.724 ms
84 bytes from 10.0.0.1 icmp_seq=2 ttl=64 time=1.288 ms

PC1-anspelov> 
```

Настройка DHCP в случае IPv4

- Просмотр статистики DHCP-сервера и выданных адресов.

```
nspelov@anspelov-gw-01:~$ show dhcp server statistics
Pool          Size    Leases    Available  Usage
-----
nspelov       252      1         251      0%
nspelov@anspelov-gw-01:~$ show dhcp server leases
P address      Hardware address    State    Lease start          Lease expiration      Remaining      Pool
-----
-----
0.0.0.2        00:50:79:66:68:00   active   2025/12/05 17:20:35  2025/12/06 17:20:35  23:58:29      ans
elov PC1-anspelov1
nspelov@anspelov-gw-01:~$
```

Настройка DHCP в случае IPv4

- Просмотр журнала работы DHCP-сервера.

```
elov@anspelov-gw-01:~$ show log | grep dhcp
05 16:58:08 dhclient-script-vyos[2404]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd
client
05 16:58:09 vyos-hostsd[700]: Request data: {"type": "search_domains", "op": "delete", "data": [{"
"eth0"}]}
05 16:58:09 dhclient-script-vyos[2404]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd
ent
05 16:58:09 vyos-hostsd[700]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dh
"eth0"}]}
05 16:59:41 dhclient-script-vyos[2438]: Deleting search domains with tag "dhcp-eth0" via vyos-hos
-client
05 16:59:42 vyos-hostsd[700]: Request data: {"type": "search_domains", "op": "delete", "data": [{"
"eth0"}]}
05 16:59:42 dhclient-script-vyos[2438]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd
ent
05 16:59:42 vyos-hostsd[700]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dh
"eth0"}]}
05 17:01:31 dhclient-script-vyos[2615]: Deleting search domains with tag "dhcp-eth0" via vyos-hos
-client
05 17:01:31 vyos-hostsd[700]: Request data: {"type": "search_domains", "op": "delete", "data": [{"
"eth0"}]}
05 17:01:31 dhclient-script-vyos[2615]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd
ent
05 17:01:31 vyos-hostsd[700]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dh
"eth0"}]}
05 17:03:14 dhclient-script-vyos[2800]: Deleting search domains with tag "dhcp-eth0" via vyos-hos
-client
05 17:03:15 vyos-hostsd[700]: Request data: {"type": "search_domains", "op": "delete", "data": [{"
"eth0"}]}
05 17:03:15 dhclient-script-vyos[2800]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd
ent
05 17:03:15 vyos-hostsd[700]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dh
"eth0"}]}
05 17:05:28 dhclient-script-vyos[3197]: Deleting search domains with tag "dhcp-eth0" via vyos-hos
-client
05 17:05:28 vyos-hostsd[700]: Request data: {"type": "search_domains", "op": "delete", "data": [{"
"eth0"}]}
05 17:05:28 dhclient-script-vyos[3197]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd
ent
05 17:05:28 vyos-hostsd[700]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dh
```

Настройка DHCP в случае IPv4

- Проанализируем захваченные анализатором трафика пакеты

The screenshot displays the Wireshark interface with a list of captured packets. The packet list pane shows several DHCP Discover and Offer packets from 10.0.0.0 to 10.0.0.1. The packet details pane for packet 62 (342 bytes) is expanded, showing the Ethernet II, Internet Protocol Version 4, and Dynamic Host Configuration Protocol (Discover) layers. The packet bytes pane shows the raw data in hexadecimal and ASCII.

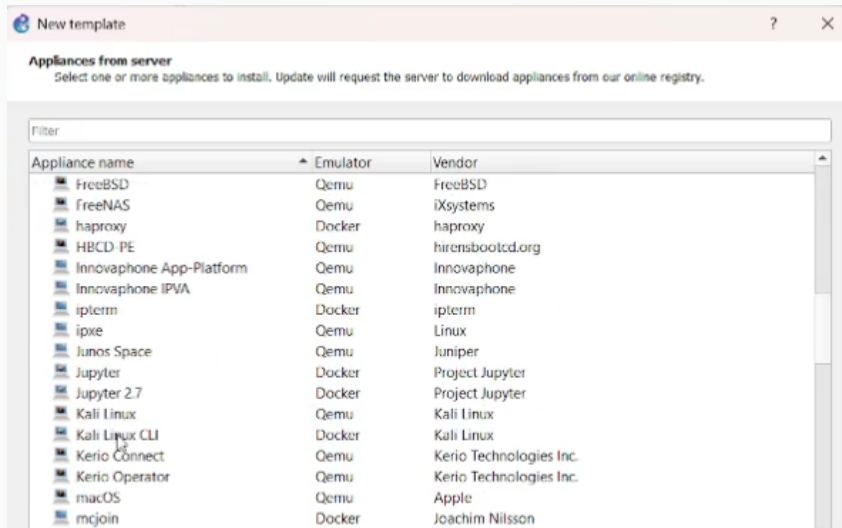
Time	Source	Destination	Protocol	Length	Info
59.1052.020543	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8feb0c67
60.1094.629967	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x21e10341
61.1099.301985	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x21e10341
62.1103.916269	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x21e10341
63.1112.355157	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x21e10341
64.1122.589111	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x21e10341
65.1531.757444	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0xbcb9ec66
66.1531.765202	0c:90:4d:13:00:00	Broadcast	ARP	42	Who has 10.0.0.2? Tell 10.0.0.1
67.1532.757322	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0xbcb9ec66
68.1532.774852	10.0.0.1	10.0.0.2	DHCP	342	DHCP Offer - Transaction ID 0xbcb9ec66
69.1532.788182	0c:90:4d:13:00:00	Broadcast	ARP	42	Who has 10.0.0.2? Tell 10.0.0.1
70.1533.812555	0c:90:4d:13:00:00	Broadcast	ARP	42	Who has 10.0.0.2? Tell 10.0.0.1
71.1535.757767	0.0.0.0	255.255.255.255	DHCP	406	DHCP Request - Transaction ID 0xbcb9ec66
72.1535.800020	10.0.0.1	10.0.0.2	DHCP	342	DHCP ACK - Transaction ID 0xbcb9ec66
73.1536.758127	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 10.0.0.2 (Request)
74.1537.758386	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 10.0.0.2 (Request)
75.1538.758745	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 10.0.0.2 (Request)
76.1588.885630	Private_66:68:00	Broadcast	ARP	64	Who has 10.0.0.1? Tell 10.0.0.2
77.1588.886142	0c:90:4d:13:00:00	Private_66:68:00	ARP	42	10.0.0.1 is at 0c:90:4d:13:00:00
78.1588.887172	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) request id=0x9814, seq=1/256, ttl=64 (reply in 79)
79.1588.894591	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) reply id=0x9814, seq=1/256, ttl=64 (request in 78)
80.1589.897010	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) request id=0x9914, seq=2/512, ttl=64 (reply in 81)
81.1589.898031	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) reply id=0x9914, seq=2/512, ttl=64 (request in 80)
82.1593.974055	0c:90:4d:13:00:00	Private_66:68:00	ARP	42	Who has 10.0.0.2? Tell 10.0.0.1
83.1593.975088	Private_66:68:00	0c:90:4d:13:00:00	ARP	42	10.0.0.2 is at 00:50:79:66:68:00

Frame 62: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface -, id 0
Ethernet II, Src: 0c:90:4d:13:00:00 (0c:90:4d:13:00:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
User Datagram Protocol, Src Port: 68, Dst Port: 67
Dynamic Host Configuration Protocol (Discover)

0000 ff ff ff ff ff ff 0c 90 4d 13 00 00 08 00 45 10
0010 01 48 00 00 00 00 00 11 39 96 00 00 00 00 ff ff
0020 ff ff 00 44 00 43 01 34 61 88 01 01 06 00 21 e1
0030 03 41 00 09 00 00 00 00 00 00 00 00 00 00 00
0040 00 00 00 00 00 00 0c 90 4d 13 00 00 00 00 00
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

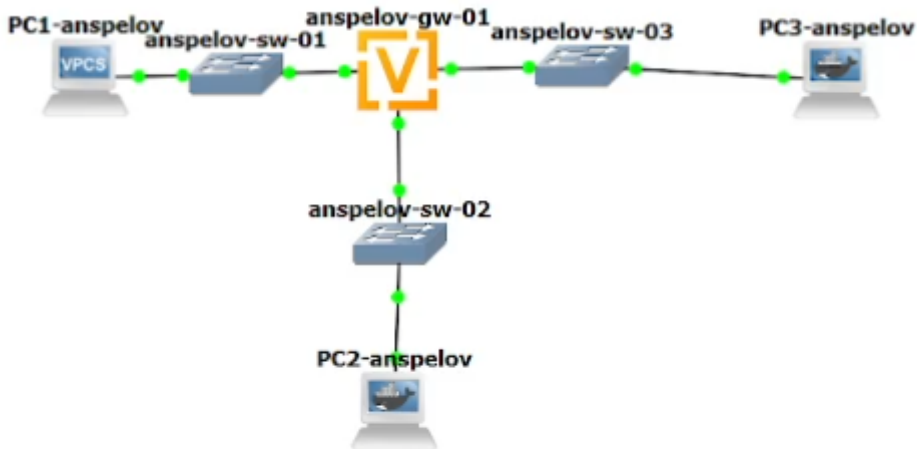
Настройка DHCP в случае IPv6

- Добавление образа Kali Linux CLI в перечень устройств в GNS3.



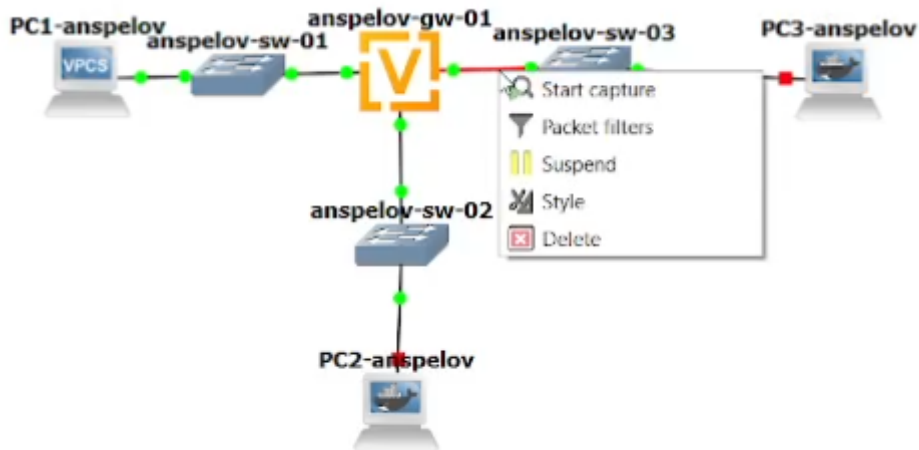
Настройка DHCP в случае IPv6

- Дополнение сети в соответствии с топологией, приведённой в лабораторной работе. Изменение отображаемых названий устройств.



Настройка DHCP в случае IPv6

- Включение захвата трафика.



Настройка DHCP в случае IPv6

- Настройка адресации IPv6 на маршрутизаторе.

```
nspelov@anspelov-gw-01:~$ configure
edit]
nspelov@anspelov-gw-01# set interfaces ethernet eth1 address 2000::1/64
edit]
nspelov@anspelov-gw-01# set interfaces ethernet eth2 address 2001::1/64
edit]
nspelov@anspelov-gw-01# show interfaces
ethernet eth0 {
    address 10.0.0.1/24
    hw-id 0c:90:4d:13:00:00
}
ethernet eth1 {
    address 2000::1/64
    hw-id 0c:90:4d:13:00:01
}
ethernet eth2 {
    address 2001::1/64
    hw-id 0c:90:4d:13:00:02
}
ethernet eth3 {
    hw-id 0c:90:4d:13:00:03
}
ethernet eth4 {
    hw-id 0c:90:4d:13:00:04
}
ethernet eth5 {
    hw-id 0c:90:4d:13:00:05
}
ethernet eth6 {
    hw-id 0c:90:4d:13:00:06
}
ethernet eth7 {
    hw-id 0c:90:4d:13:00:07
}
ethernet eth8 {
    hw-id 0c:90:4d:13:00:08
}
ethernet eth9 {
    hw-id 0c:90:4d:13:00:09
}
loopback lo {
```

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Настройка DHCP в случае IPv6

- Настройка на маршрутизаторе DHCPv6 без отслеживания состояния.
Добавление конфигурации DHCP-сервера.

```
nspelev@anspelov-gw-01# set service router-advert interface eth1 prefix 2000::/
edit]
nspelev@anspelov-gw-01# set service router-advert interface eth1 other-config-f
edit]
nspelev@anspelov-gw-01# commit
edit]
nspelev@anspelov-gw-01# save
aving configuration to '/config/config.boot'...
one
edit]
nspelev@anspelov-gw-01# set service dhcpv6-server shared-network-name anspelev-
edit]
nspelev subnet 2000::0/64t service dhcpv6-server shared-network-name anspelev-
edit]
edit]
nspelev common-options name-server 2000::1server shared-network-name anspelev-
edit]
nspelev common-options domain-search anspelev.nethared-network-name anspelev-
edit]
nspelev@anspelov-gw-01# commit
edit]
nspelev@anspelov-gw-01# save
aving configuration to '/config/config.boot'...
one
edit]
nspelev@anspelov-gw-01# run show configuration
nterfaces {
  ethernet eth0 {
    address 10.0.0.1/24
    hw-id 0c:90:4d:13:00:00
  }
  ethernet eth1 {
    address 2000::1/64
    hw-id 0c:90:4d:13:00:01
  }
  ethernet eth2 {
    address 2001::1/64
    hw-id 0c:90:4d:13:00:02
  }
  ethernet eth3 {
```

Настройка DHCP в случае IPv6

- Проверка на узле PC2-anspelov настройки сети.

```
PC2-anspelov - PuTTY
inet6 addr: 2000::42:26ff:fec7:e700/64 Scope:Global
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:11 errors:0 dropped:0 overruns:0 frame:0
TX packets:12 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:1306 (1.2 KiB) TX bytes:952 (932.0 B)

eth1    Link encap:Ethernet HWaddr 02:42:26:C7:E7:01
        inet6 addr: fe80::42:26ff:fec7:e701/64 Scope:Link
        UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

lo      Link encap:Local Loopback
        inet addr:127.0.0.1 Mask:255.0.0.0
        inet6 addr: ::1/128 Scope:Host
        UP LOOPBACK RUNNING MTU:65536 Metric:1
        RX packets:32 errors:0 dropped:0 overruns:0 frame:0
        TX packets:32 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:2512 (2.4 KiB) TX bytes:2512 (2.4 KiB)

/ # route -n -A inet6
Kernel IPv6 routing table
Destination                Next Hop                    Flags Metric R
eF    Use Iface
2000::/64
    0 eth0                  ::                          UA    256    1
fe80::/64
    0 eth0                  ::                          U     256    1
fe80::/64
    0 eth1                  ::                          U     256    1
::/0
    0 eth0                  fe80::e90:4dff:fe13:1      UGDA  1024   1
::1/128
    0 lo                     ::                          Un     0     3
2000::42:26ff:fec7:e700/128
    0 eth0                  ::                          Un     0     2
fe80::42:26ff:fec7:e700/128
    0 eth0                  ::                          Un     0     2
fe80::42:26ff:fec7:e701/128
    0 eth1                  ::                          Un     0     3
fe80::42:26ff:fec7:e701/128
    0 eth1                  ::                          Un     0     3
```

Настройка DHCP в случае IPv6

- Пинг маршрутизатора на узле PC2-anspelov, проверка настройки DNS, получение адреса по DHCPv6.

```
/ # ping 2000::1 -c 2
PING 2000::1 (2000::1): 56 data bytes
64 bytes from 2000::1: seq=0 ttl=64 time=2.739 ms
64 bytes from 2000::1: seq=1 ttl=64 time=1.196 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 1.196/1.967/2.739 ms
/ # cat /etc/resolv.conf
/ #
/ #
/ # dhclient -6 -S -v eth0
```

- Пинг маршрутизатора от узла PC2-anspelov и проверка настройки DNS.

```
/ # ping 2000::1 -c2
PING 2000::1 (2000::1): 56 data bytes
64 bytes from 2000::1: seq=0 ttl=64 time=2.181 ms
64 bytes from 2000::1: seq=1 ttl=64 time=1.031 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 1.031/1.606/2.181 ms
/ # cat /etc/resolv.conf
```

Настройка DHCP в случае IPv6

- Просмотр статистики на маршрутизаторе DHCP-сервера и выданных адресов.

```
nspelov@anspelov-gw-01# run show dhcpv6 server leases
Pv6 address      State      Last communication      Lease expiration      Remaining      Type      Pool
IAID_DUID
-----
edit]
```

Настройка DHCP в случае IPv6

- Проанализируем захваченные анализатором трафика пакеты

Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка					
Примените фильтр отображения ... <Ctrl-/>					
Time	Source	Destination	Protocol	Length	Info
1 0.000000	::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
2 0.846175	::	ff02::1:ffc7:e700	ICMPv6	86	Neighbor Solicitation for fe80::42:26ff:fec7:e700
3 0.910273	::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
4 1.872834	fe80::42:26ff:fec7::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
5 1.873871	fe80::42:26ff:fec7::	ff02::2	ICMPv6	70	Router Solicitation from 02:42:26:c7:e7:00
6 2.254183	fe80::42:26ff:fec7::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
7 6.286791	fe80::42:26ff:fec7::	ff02::2	ICMPv6	70	Router Solicitation from 02:42:26:c7:e7:00
8 14.990687	fe80::42:26ff:fec7::	ff02::2	ICMPv6	70	Router Solicitation from 02:42:26:c7:e7:00
9 33.422626	fe80::42:26ff:fec7::	ff02::2	ICMPv6	70	Router Solicitation from 02:42:26:c7:e7:00
10 70.285540	fe80::42:26ff:fec7::	ff02::2	ICMPv6	70	Router Solicitation from 02:42:26:c7:e7:00
11 137.196878	fe80::e90:4dff:fe13::	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
12 137.204562	fe80::e90:4dff:fe13::	ff02::16	ICMPv6	170	Multicast Listener Report Message v2
13 137.260429	fe80::e90:4dff:fe13::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
14 137.453066	::	ff02::1:ff00:1	ICMPv6	86	Neighbor Solicitation for 2000::1
15 139.916110	fe80::42:26ff:fec7::	ff02::2	ICMPv6	70	Router Solicitation from 02:42:26:c7:e7:00
16 251.997378	fe80::e90:4dff:fe13::	ff02::1	ICMPv6	118	Router Advertisement from 0c:90:4d:13:00:01
17 252.186033	::	ff02::1:ffc7:e700	ICMPv6	86	Neighbor Solicitation for 2000::42:26ff:fec7:e700
18 268.019126	fe80::e90:4dff:fe13::	ff02::1	ICMPv6	118	Router Advertisement from 0c:90:4d:13:00:01
19 284.036201	fe80::e90:4dff:fe13::	ff02::1	ICMPv6	118	Router Advertisement from 0c:90:4d:13:00:01
20 492.426455	fe80::e90:4dff:fe13::	ff02::16	ICMPv6	110	Multicast Listener Report Message v2
21 493.365753	fe80::e90:4dff:fe13::	ff02::16	ICMPv6	110	Multicast Listener Report Message v2
22 674.011594	fe80::e90:4dff:fe13::	ff02::1	ICMPv6	118	Router Advertisement from 0c:90:4d:13:00:01
23 1242.676591	fe80::e90:4dff:fe13::	ff02::1	ICMPv6	118	Router Advertisement from 0c:90:4d:13:00:01
24 1386.694450	2000::42:26ff:fec7::	ff02::1:ff00:1	ICMPv6	86	Neighbor Solicitation for 2000::1 from 02:42:26:c7:e7:00
25 1386.695987	2000::1	2000::42:26ff:fec7::	ICMPv6	86	Neighbor Advertisement 2000::1 (rtr, sol, ovr) is at 0c:90:4d:13:00:01
26 1386.695987	2000::42:26ff:fec7::	2000::1	ICMPv6	118	Echo (ping) request id=0x5501, seq=0, hop limit=64 (reply in 27)
Name 17: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface -, id 0					
Ethernet II, Src: 02:42:26:c7:e7:00 (02:42:26:c7:e7:00), Dst: IPv6mcast_ff:c7:e7:00 (33:33:ff:c7:e7:00)					
Internet Protocol Version 6, Src: ::, Dst: ff02::1:ffc7:e700					
Internet Control Message Protocol v6					

Настройка DHCP в случае IPv6

- Настройка на маршрутизаторе DHCPv6 с отслеживанием состояния. Настройка на интерфейсе eth2 маршрутизатора объявления о маршрутизаторах. Добавление конфигурации DHCP-сервера на маршрутизатор.

```
nspelov@anspelov-gw-01# set service router-advert interface eth2 managed-flag
edit]
nspelov@anspelov-gw-01# commit
edit]
nspelov@anspelov-gw-01# save
aving configuration to '/config/config.boot'...
one
edit]
tateful@anspelov-gw-01# set service dhcpv6-server shared-network-name anspelev-
edit]
tateful subnet 2001::0/64et service dhcpv6-server shared-network-name anspelev-
edit]
tateful subnet 2001::0/64 name-server 2001::1rver shared-network-name anspelev-
edit]
tateful subnet 2001::0/64 domain-search anspelev.netared-network-name anspelev-
edit]
tateful subnet 2001::0/64 address-range start 2001::100 stop 2001::199anspelev-
```

- Просмотр статистики DHCP-сервера на маршрутизаторе и выданных адресов.

```
nspelov@anspelov-gw-01# run show dhcpv6 server leases
Pv6 address      State      Last communication    Lease expiration      Remaining      Type      Pool
IAID_DUID
-----
edit]
```

Настройка DHCP в случае IPv6

- Подключение к узлу PC3-anspelov и проверка настройки сети.

```
PC3-anspelov - PuTTY

eth1    Link encap:Ethernet  HWaddr 02:42:80:2C:D2:01
        inet6 addr: fe80::42:80ff:fe2c:d201/64 Scope:Link
        UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

lo       Link encap:Local Loopback
        inet addr:127.0.0.1  Mask:255.0.0.0
        inet6 addr: ::1/128 Scope:Host
        UP LOOPBACK RUNNING  MTU:65536  Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

/ # route -n -A inet6
Kernel IPv6 routing table

```

Destination	Metric	Ref	Use	Iface	Next Hop	Flags
fe80::/64	256	1		0 eth0	::	U
fe80::/64	256	1		0 eth1	::	U
::/0	1024	1		0 eth0	fe80::e90:4dff:fe13:2	UGDA
::1/128	0	2		0 lo	::	Un
fe80::42:80ff:fe2c:d200/128	0	2		0 eth0	::	Un
fe80::42:80ff:fe2c:d201/128	0	3		0 eth1	::	Un
ff00::/8	256	3		0 eth0	::	U

- Проверка на узле PC3-anspelov настройки DNS и получение адреса по DHCPv6.

```
/ # cat /etc/resolv.conf  
/ #
```

Настройка DHCP в случае IPv6

- Проверка на узле PC3-anspelov настройки сети, пинг маршрутизатора и проверка настройки.

```
PC3-anspelov - PuTTY
lo
  Link encap:Local Loopback
  inet addr:127.0.0.1  Mask:255.0.0.0
  inet6 addr: ::1/128 Scope:Host
  UP LOOPBACK RUNNING  MTU:65536  Metric:1
  RX packets:0 errors:0 dropped:0 overruns:0 frame:0
  TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
  collisions:0 txqueuelen:1000
  RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

# route -n -A inet6
Kernel IPv6 routing table

```

Destination	Metric Ref	Use Iface	Next Hop	Flags
fe80::/64	1	0 eth0	::	U
fe80::/64	1	0 eth1	::	U
::/0	1	0 eth0	fe80::e90:4dff:fe13:2	UGDA
::1/128	2	0 lo	::	Un
fe80::42:80ff:fe2c:d200/128	2	0 eth0	::	Un
fe80::42:80ff:fe2c:d201/128	3	0 eth1	::	Un
ff00::/8	3	0 eth0	::	U
ff00::/8	1	0 eth1	::	U
::/0	1	0 lo	::	!n

- Просмотр на маршрутизаторе статистики DHCP-сервера и выданных адресов.

```
nspelov@anspelov-gw-01# run show dhcpv6 server leases
Pv6 address      State      Last communication      Lease expiration      Remaining      Type      Pool
IAID_DUID
-----
edit]
```

Настройка DHCP в случае IPv6

- Проанализируем захваченные анализатором трафика пакеты

Time	Source	Destination	Protocol	Length	Info
1 0.000000	::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
2 0.395594	::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
3 0.939418	::	ff02::1:ff2c:d200	ICMPv6	86	Neighbor Solicitation for fe80::42:80ff:fe2c:d200
4 1.963348	fe80::42:80ff:fe2c::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
5 1.963860	fe80::42:80ff:fe2c::	ff02::2	ICMPv6	70	Router Solicitation from 02:42:80:2c:d2:00
6 2.923419	fe80::42:80ff:fe2c::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
7 6.027620	fe80::42:80ff:fe2c::	ff02::2	ICMPv6	70	Router Solicitation from 02:42:80:2c:d2:00
8 13.963631	fe80::42:80ff:fe2c::	ff02::2	ICMPv6	70	Router Solicitation from 02:42:80:2c:d2:00
9 29.581058	fe80::42:80ff:fe2c::	ff02::2	ICMPv6	70	Router Solicitation from 02:42:80:2c:d2:00
10 62.090385	fe80::42:80ff:fe2c::	ff02::2	ICMPv6	70	Router Solicitation from 02:42:80:2c:d2:00
11 123.529260	fe80::42:80ff:fe2c::	ff02::2	ICMPv6	70	Router Solicitation from 02:42:80:2c:d2:00
12 141.317269	fe80::e90:4dff:fe13::	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
13 141.322930	fe80::e90:4dff:fe13::	ff02::16	ICMPv6	170	Multicast Listener Report Message v2
14 142.123408	fe80::e90:4dff:fe13::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
15 142.123408	::	ff02::1:ff00:1	ICMPv6	86	Neighbor Solicitation for 2001::1
16 244.361294	fe80::42:80ff:fe2c::	ff02::2	ICMPv6	70	Router Solicitation from 02:42:80:2c:d2:00
17 494.219294	fe80::42:80ff:fe2c::	ff02::2	ICMPv6	70	Router Solicitation from 02:42:80:2c:d2:00
18 985.744277	fe80::42:80ff:fe2c::	ff02::2	ICMPv6	70	Router Solicitation from 02:42:80:2c:d2:00
19 1597.922320	fe80::e90:4dff:fe13::	ff02::1	ICMPv6	86	Router Advertisement from 0c:90:4d:13:00:02
20 1613.951544	fe80::e90:4dff:fe13::	ff02::1	ICMPv6	86	Router Advertisement from 0c:90:4d:13:00:02
21 1629.968808	fe80::e90:4dff:fe13::	ff02::1	ICMPv6	86	Router Advertisement from 0c:90:4d:13:00:02
22 1820.372660	fe80::e90:4dff:fe13::	ff02::16	ICMPv6	110	Multicast Listener Report Message v2
23 1820.547015	fe80::e90:4dff:fe13::	ff02::16	ICMPv6	110	Multicast Listener Report Message v2
24 1914.286293	fe80::42:80ff:fe2c::	2001::1	ICMPv6	118	Echo (ping) request id=0x3201, seq=0, hop limit=64 (reply in 25)
25 1914.289395	2001::1	fe80::42:80ff:fe2c::	ICMPv6	118	Echo (ping) reply id=0x3201, seq=0, hop limit=64 (request in 24)
26 1915.285966	fe80::42:80ff:fe2c::	2001::1	ICMPv6	118	Echo (ping) request id=0x3201, seq=1, hop limit=64 (reply in 27)

Frame 2: 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on interface -, id 0
Ethernet II, Src: 02:42:80:2c:d2:00 (02:42:80:2c:d2:00), Dst: IPv6mcast_16 (33:33:00:00:00:16)
Internet Protocol Version 6, Src: ::, Dst: ff02::16
Internet Control Message Protocol v6

0000 33 33 00 00 00 16 02 42 00 2c d2 00 00 86 dd 60 00
0010 00 00 00 24 00 01 00 00 00 00 00 00 00 00 00 00
0020 00 00 00 00 00 00 ff 02 00 00 00 00 01 00 00 00
0030 00 00 00 00 00 16 3a 00 05 02 00 00 01 00 8f 00
0040 9d 5d 00 00 00 01 04 00 00 00 ff 02 00 00 00 00

Вывод

- В ходе выполнения лабораторной работы мы получили навыки настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.